

GEOS-MPM Validation Suite Report

Generated by validation/runValidationSuite

May 17, 2026

Running and rerunning

Cases are staged under testRunDirectory/validation/<case>. The suite records failures and continues unless --stop-on-error is supplied.

```
cd scripts/preProcessing/materialPointMethodParticleFileWriter/validation
./runValidationSuite all --python /usr/tce/bin/python3 --force
./runValidationSuite report --python /usr/tce/bin/python3
```

Timing and dispatch summary

Case	Status	Family	GEOS job	GEOS state	GEOS wall	Po
geomechanics__geoOHCrD_2024_ISR08_M5run	not present	Geomechanics				
geomechanics__geomechanicsHydrostaticCreep_HP03	not present	Geomechanics				
geomechanics__geomechanicsTriaxialCompression_ISR03	not present	Geomechanics				
geomechanics__geomechanicsTriaxialCompression_TP1	not present	Geomechanics				
geomechanics__geomechanicsTriaxialCompression_TP2	not present	Geomechanics				
geomechanics__geomechanicsTriaxialCompression_TP3	not present	Geomechanics				
geomechanics__geomechanicsTriaxialCompression_TP3	not present	Geomechanics				
geomechanics__geomechanicsTriaxialCompression_TP3	not present	Geomechanics				
polymer__polymerCompression	submitted	Polymer validation	6046423	COMPLETED	00:00:29	00:
polymer__polymerTension	submitted	Polymer validation	6046425	COMPLETED	00:00:34	00:

geomechanics / geoOHCrD 2024 ISR08 V5

Validation case for Geomechanics.

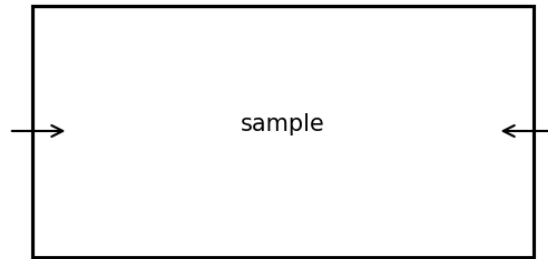


Figure 1: Schematic for geomechanics_geoOHCrD_2024_ISR08_V5.

Code features exercised.

- reaction-history output
- box-average history
- Geomechanics constitutive model
- temperature table/event

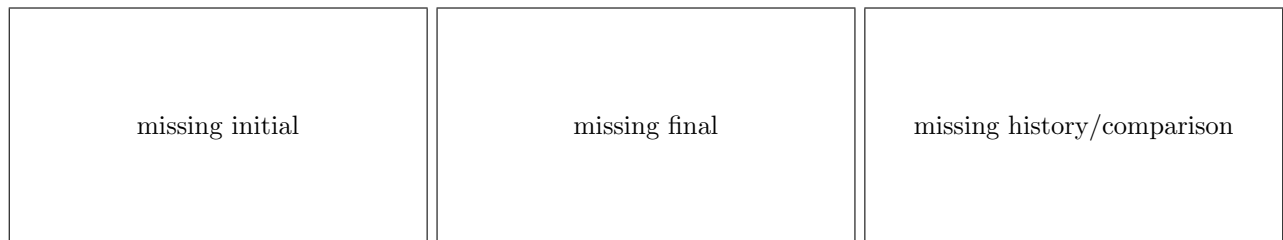


Figure 2: Initial state, final state, and history/comparison plot for geomechanics_geoOHCrD_2024_ISR08_V5.

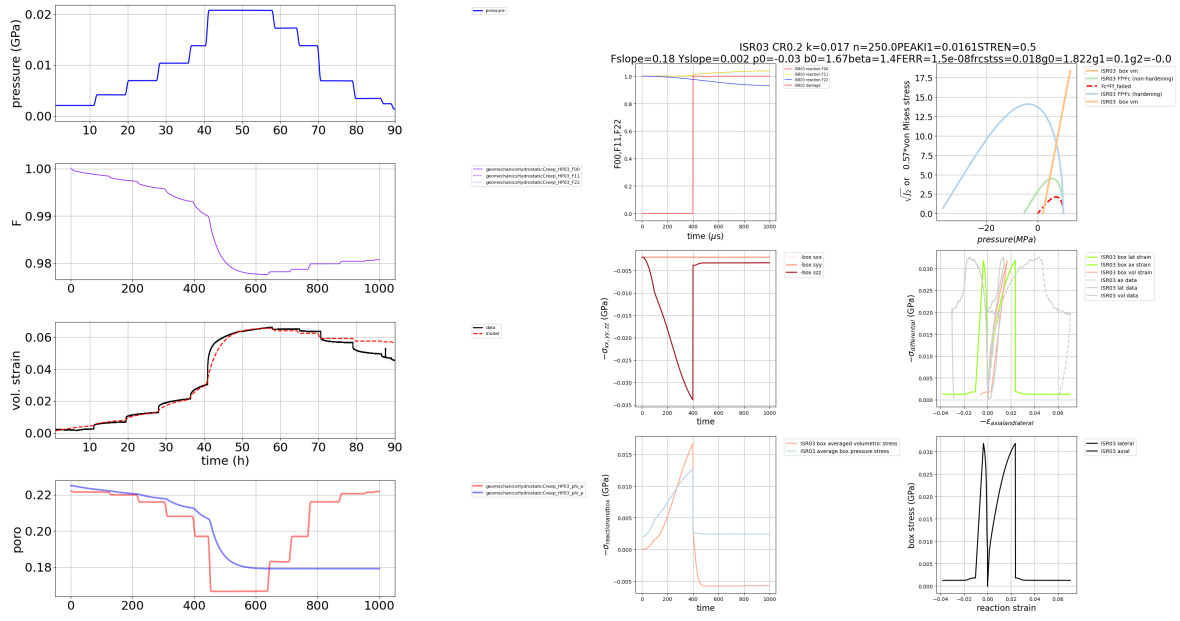


Figure 3: Existing comparison assets in the source case directory.

Reference or legacy comparison assets.

geomechanics / geomechanicsHydrostaticCreep HP03

Validation case for Geomechanics.

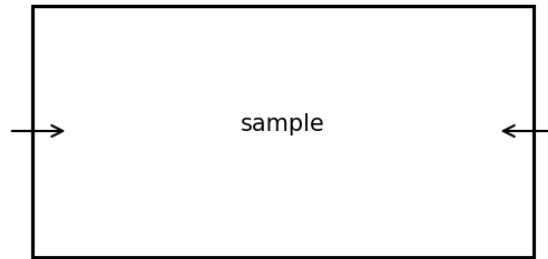


Figure 4: Schematic for geomechanics_geomechanicsHydrostaticCreep_HP03.

Code features exercised.

- reaction-history output
- box-average history
- Geomechanics constitutive model
- VonMisesJ plasticity

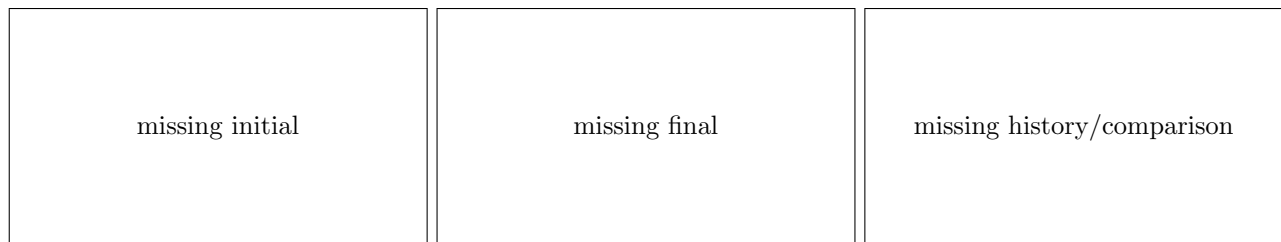


Figure 5: Initial state, final state, and history/comparison plot for geomechanics_geomechanicsHydrostaticCreep_HP03.

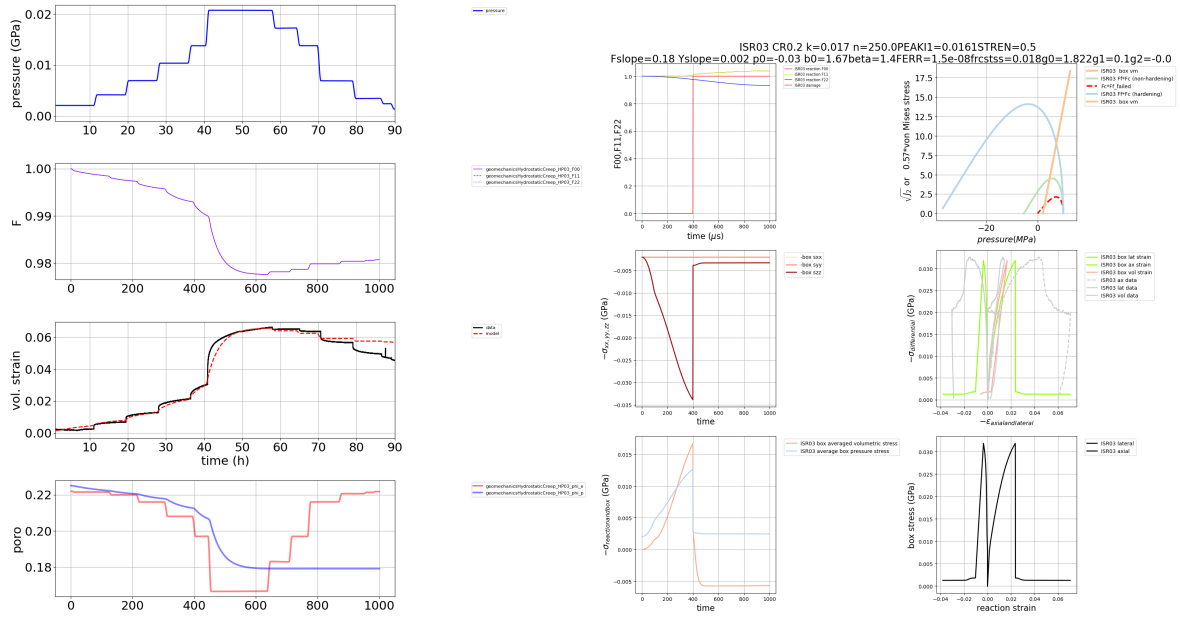


Figure 6: Existing comparison assets in the source case directory.

Reference or legacy comparison assets.

geomechanics / geomechanicsTriaxialCompression ISR03

Validation case for Geomechanics.

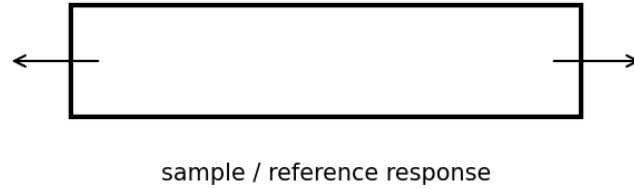


Figure 7: Schematic for geomechanics__geomechanicsTriaxialCompression_ISR03.

Code features exercised.

- FMPM update
- XPIC/PIC-family transfer
- reaction-history output
- box-average history
- Geomechanics constitutive model

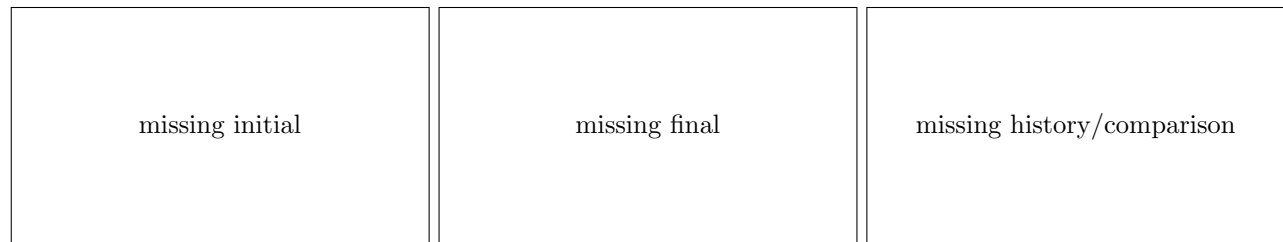


Figure 8: Initial state, final state, and history/comparison plot for geomechanics__geomechanicsTriaxialCompression_ISR03.

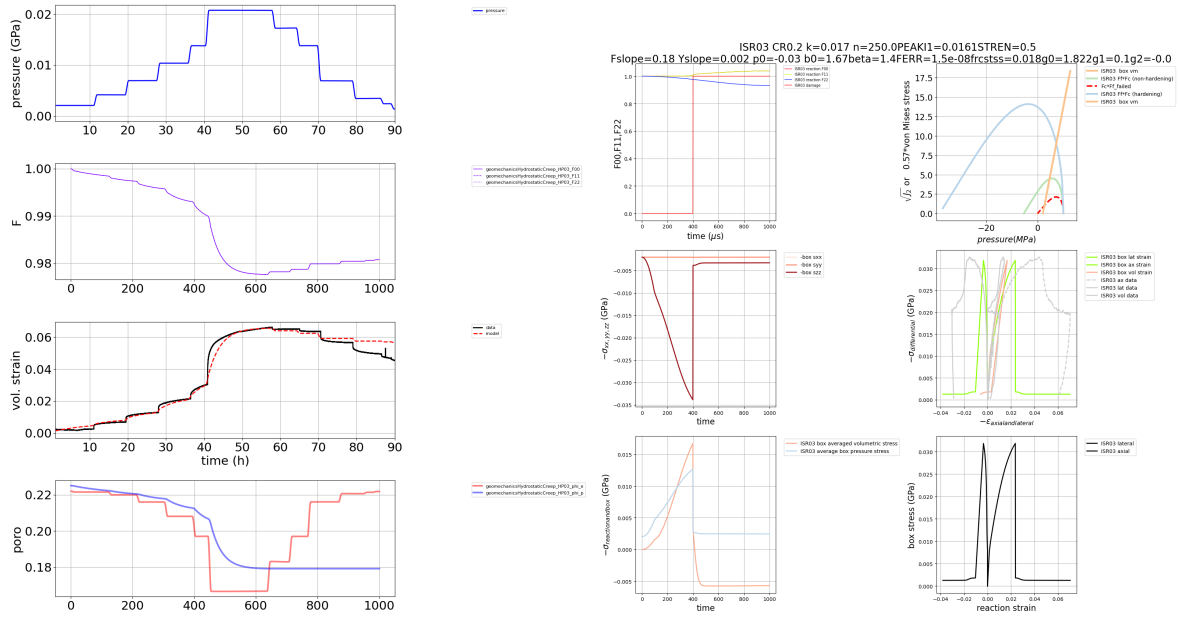


Figure 9: Existing comparison assets in the source case directory.

Reference or legacy comparison assets.

geomechanics / geomechanicsTriaxialCompression TP1

Validation case for Geomechanics.

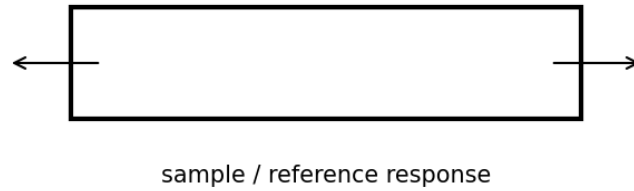


Figure 10: Schematic for geomechanics_geomechanicsTriaxialCompression_TP1.

Code features exercised.

- FMPM update
- XPIC/PIC-family transfer
- reaction-history output
- box-average history
- Geomechanics constitutive model

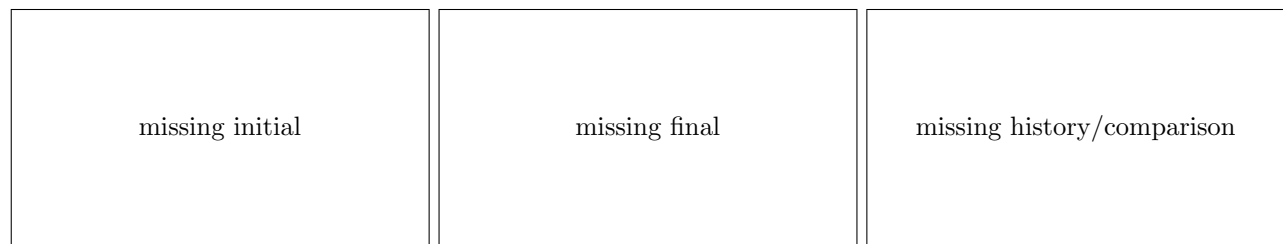


Figure 11: Initial state, final state, and history/comparison plot for geomechanics_geomechanicsTriaxialCompression_TP1.

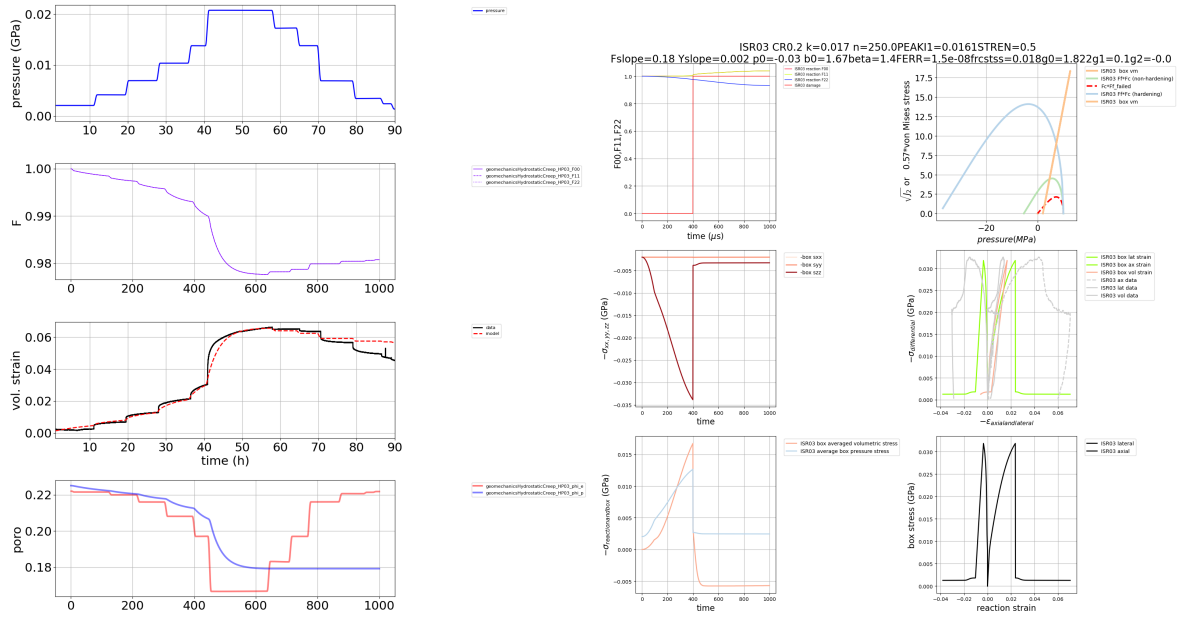


Figure 12: Existing comparison assets in the source case directory.

Reference or legacy comparison assets.

geomechanics / geomechanicsTriaxialCompression TP2

Validation case for Geomechanics.

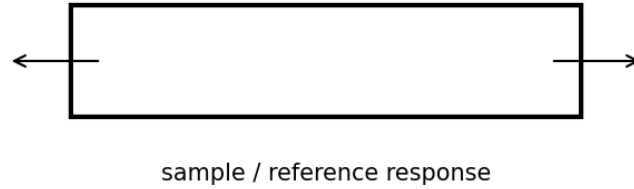


Figure 13: Schematic for geomechanics_geomechanicsTriaxialCompression_TP2.

Code features exercised.

- FMPM update
- XPIC/PIC-family transfer
- reaction-history output
- box-average history
- Geomechanics constitutive model

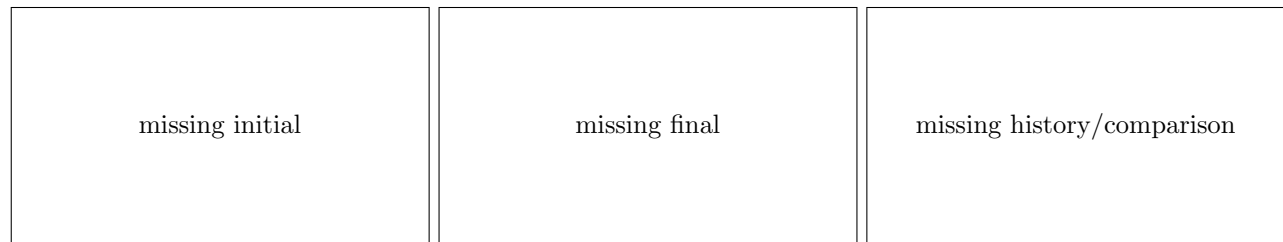


Figure 14: Initial state, final state, and history/plot for geomechanics_geomechanicsTriaxialCompression_TP2.

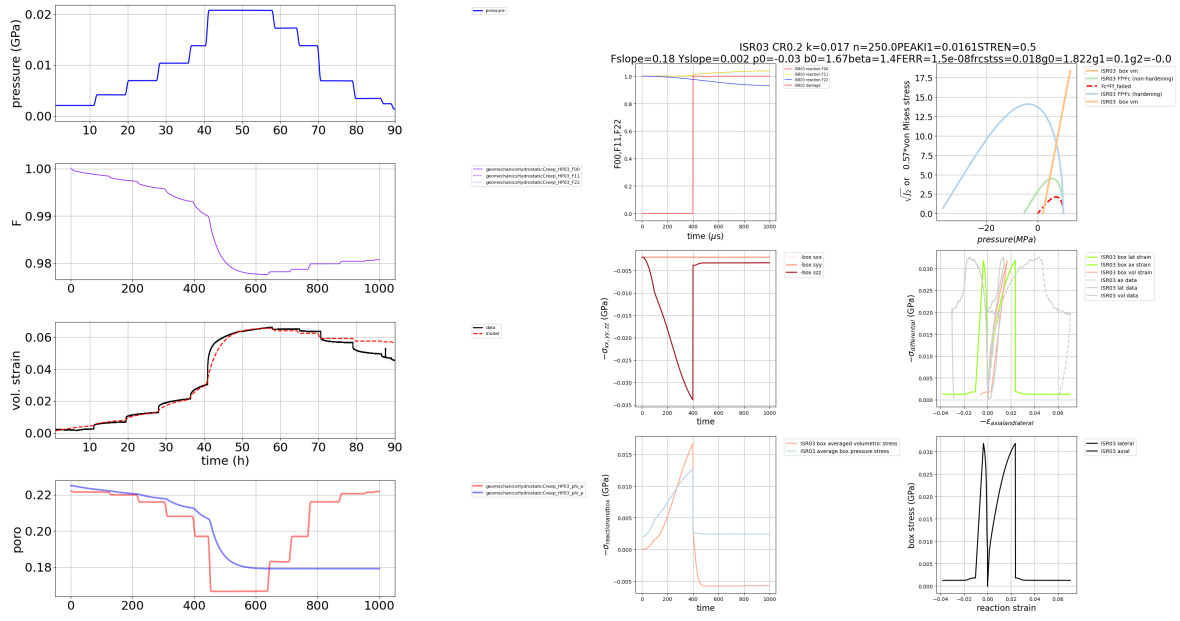


Figure 15: Existing comparison assets in the source case directory.

Reference or legacy comparison assets.

geomechanics / geomechanicsTriaxialCompression TP3

Validation case for Geomechanics.

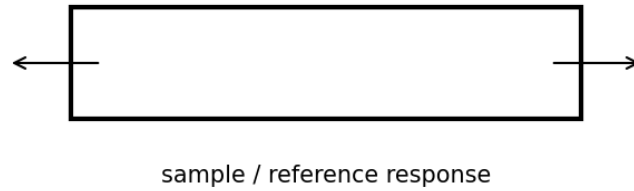


Figure 16: Schematic for geomechanics_geomechanicsTriaxialCompression_TP3.

Code features exercised.

- FMPM update
- XPIC/PIC-family transfer
- reaction-history output
- box-average history
- Geomechanics constitutive model

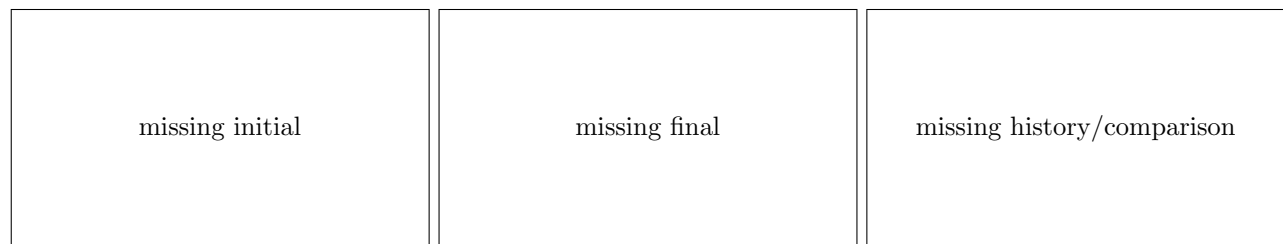


Figure 17: Initial state, final state, and history/comparison plot for geomechanics_geomechanicsTriaxialCompression_TP3.

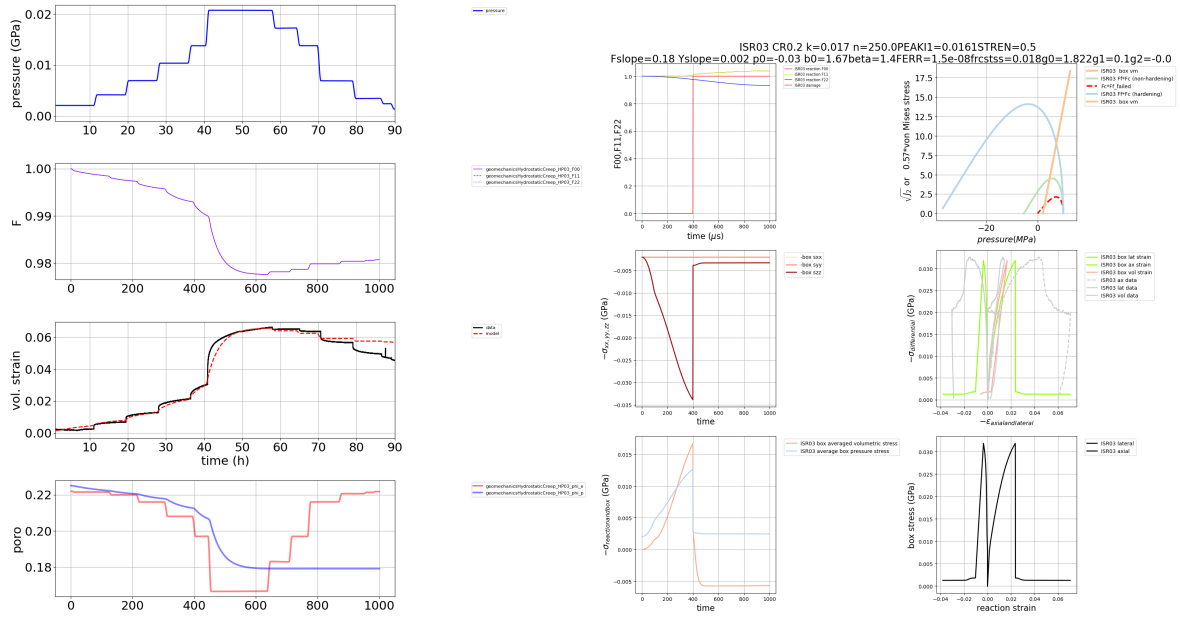


Figure 18: Existing comparison assets in the source case directory.

Reference or legacy comparison assets.

geomechanics / geomechanicsTriaxialCreep TC02

Validation case for Geomechanics.

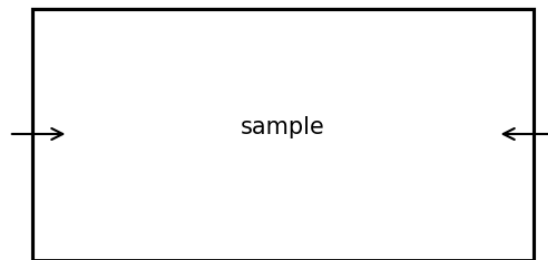


Figure 19: Schematic for geomechanics_geomechanicsTriaxialCreep_TC02.

Code features exercised.

- FMPM update
- XPIC/PIC-family transfer
- reaction-history output
- box-average history
- Geomechanics constitutive model
- VonMisesJ plasticity

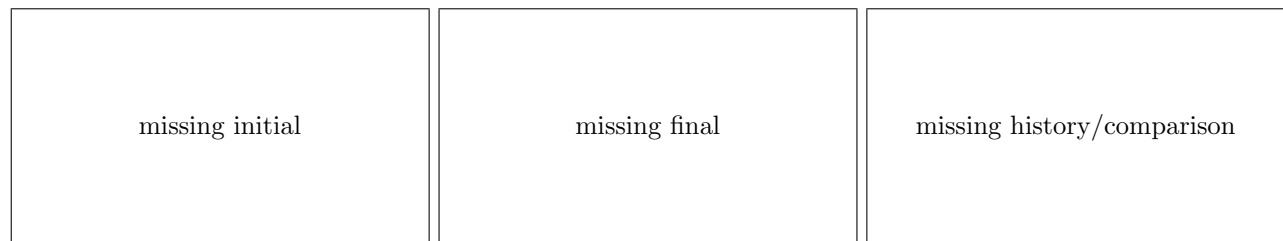


Figure 20: Initial state, final state, and history/comparison plot for geomechanics_geomechanicsTriaxialCreep_TC02.

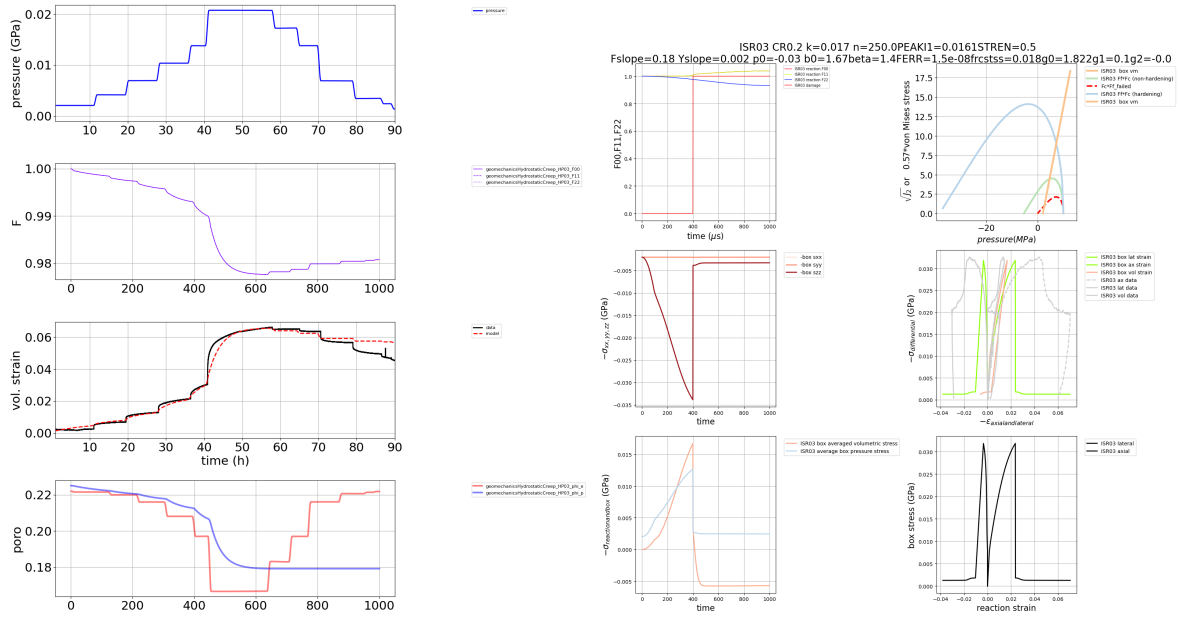


Figure 21: Existing comparison assets in the source case directory.

Reference or legacy comparison assets.

polymer / polymerCompression

Validation case for Polymer validation.

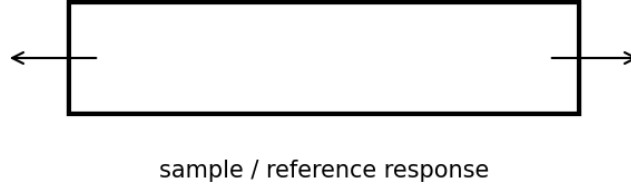


Figure 22: Schematic for polymer__polymerCompression.

Code features exercised.

- reaction-history output
- box-average history

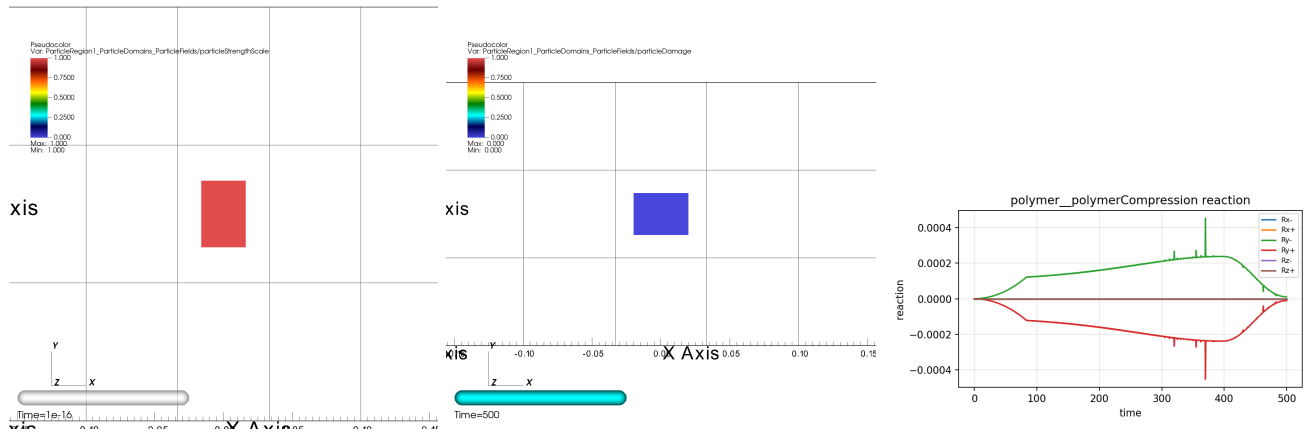


Figure 23: Initial state, final state, and history/comparison plot for polymer__polymerCompression.

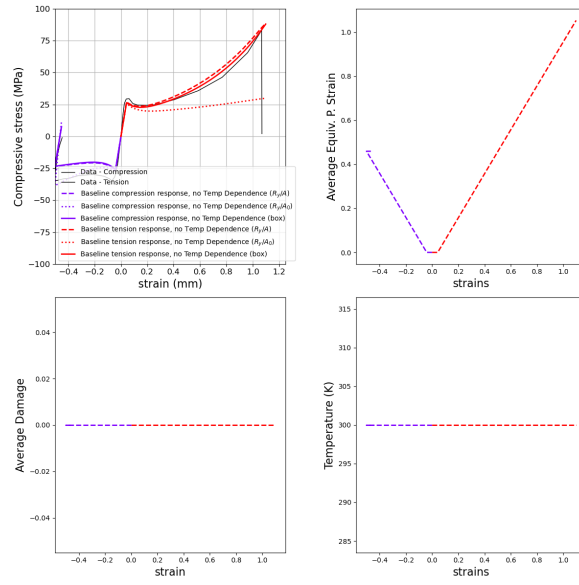


Figure 24: Existing comparison assets in the source case directory.

Reference or legacy comparison assets.

polymer / polymerTension

Validation case for Polymer validation.

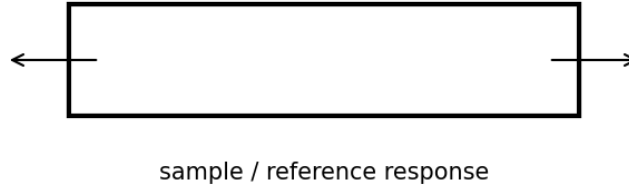


Figure 25: Schematic for polymer__polymerTension.

Code features exercised.

- reaction-history output
- box-average history

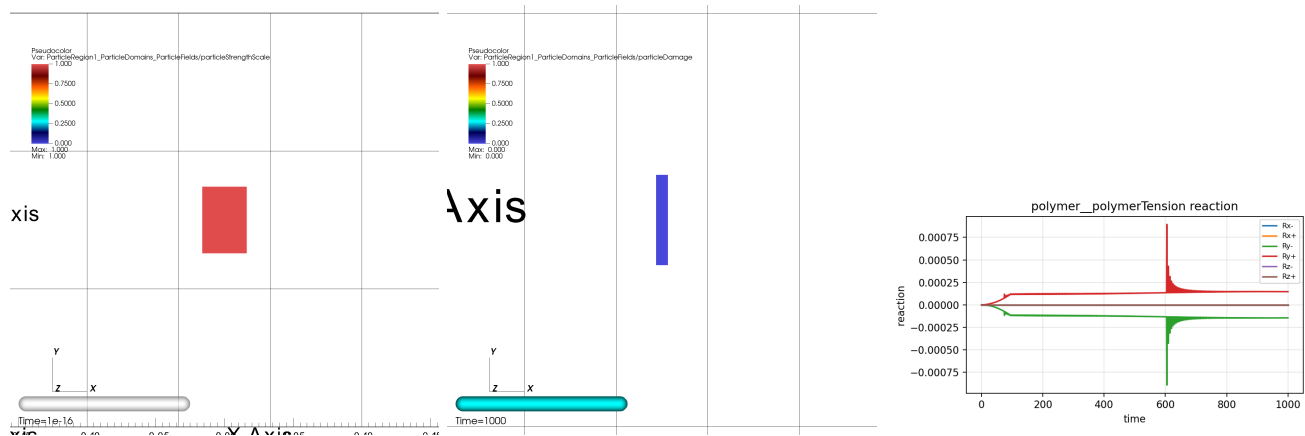


Figure 26: Initial state, final state, and history/comparison plot for polymer__polymerTension.

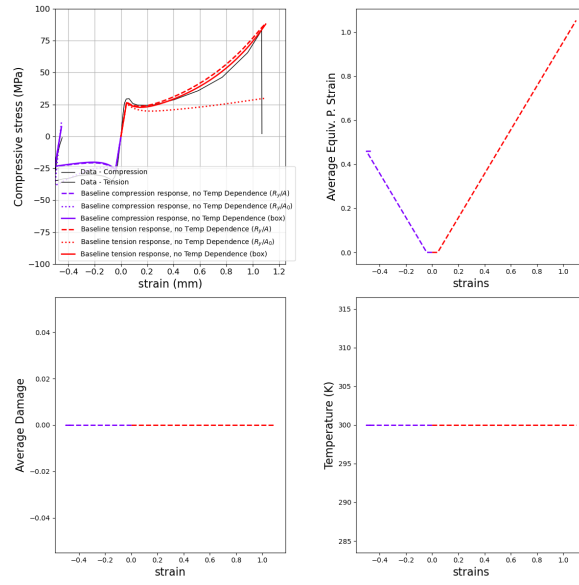


Figure 27: Existing comparison assets in the source case directory.

Reference or legacy comparison assets.